

HERMES System Installation Guide

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Abstract

High-Frequency Emergency and Rural Multimedia Exchange System (HERMES) installation procedure for the sBitx radio.

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1 Introduction

This guide goes through the steps needed for HERMES system installation.

2 Power supply and consumption

In order to connect the radio to the power source, a power cable is provided. One side of the cable comes with a XT60 connector (Figure 2) and connects to the radio, and the other end is bare ended, and should be connected to the positive (Red) and negative (Black) of a 12V battery system, or power supply. Power consumption is not more than 1.5 A in receive mode, and 9 A in transmit mode.

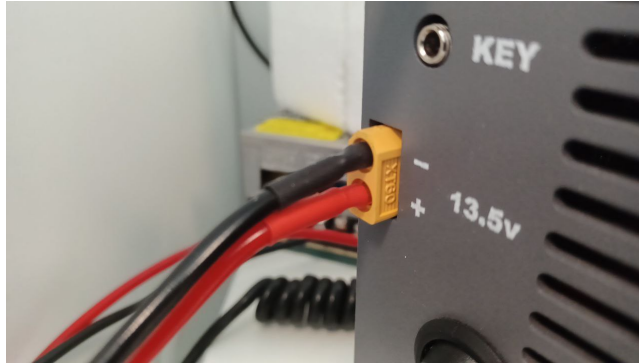


Figure 1: The radio-side of the power cables where the XT60 connector is located.

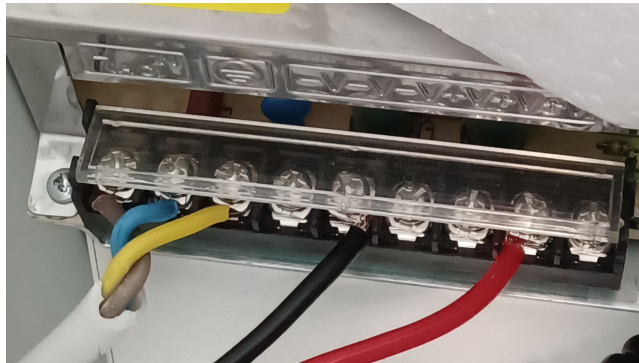


Figure 2: Red (+) and black (-) cables connected to appropriate power supply terminals.

3 Initial setup to allow boot from USB pendrive

This step is needed in order to enable the HERMES installer to run from a USB pendrive.

First connect a USB keyboard to the radio, and then connect the radio to the network (RJ45) cable with appropriate Internet connection. Then click in the Terminal icon shown in Figure 3.

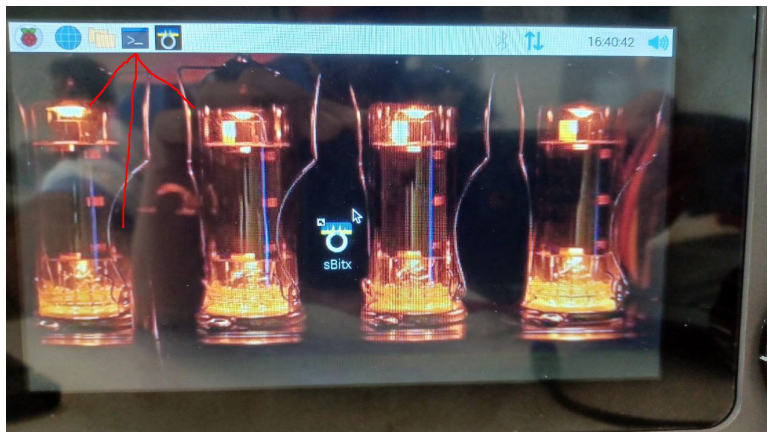


Figure 3: Screen of the radio and the "Terminal" icon pointed by a red arrow.

After opening the terminal window, type:

- `sudo su`
- `apt-get update`
- `apt-get install rpi-eeeprom`
- `raspi-config`

At this point the screen shown in Figure 4 should appear.

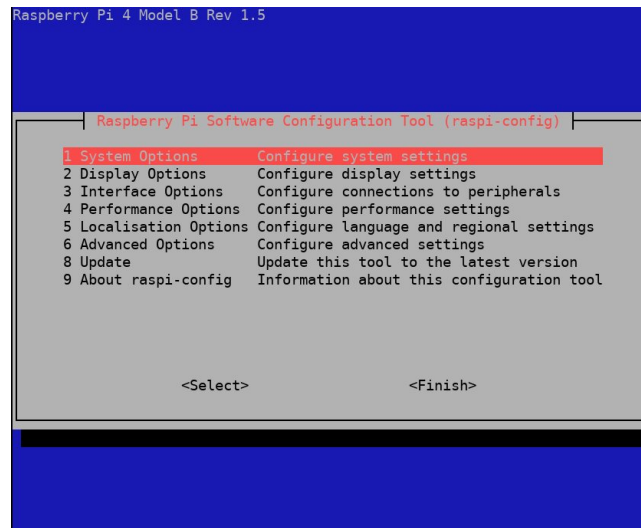


Figure 4: Configuration software screen (raspi-config tool).

Choose the options of the menu in the following order using the keyboard arrow and Tab keys (as shows in Figures):

- Step 1: Option 6 (Advanced Options)
- Step 2: Option A6 (Boot Order)
- Step 3: Option B2 (USB Boot)
- Step 4: Finish

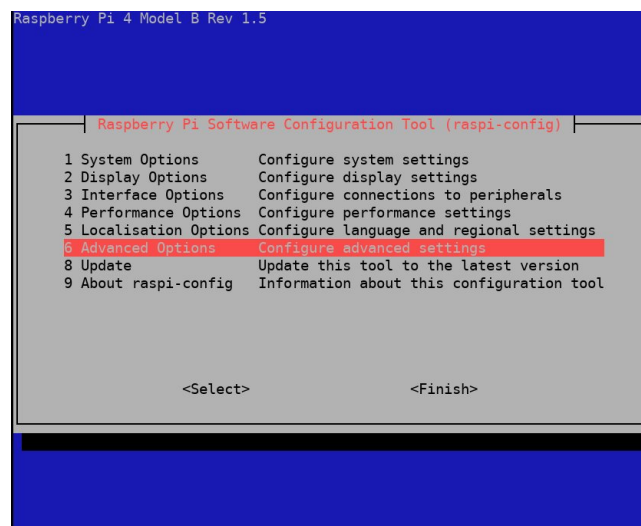


Figure 5: Menu option to select (step 1).

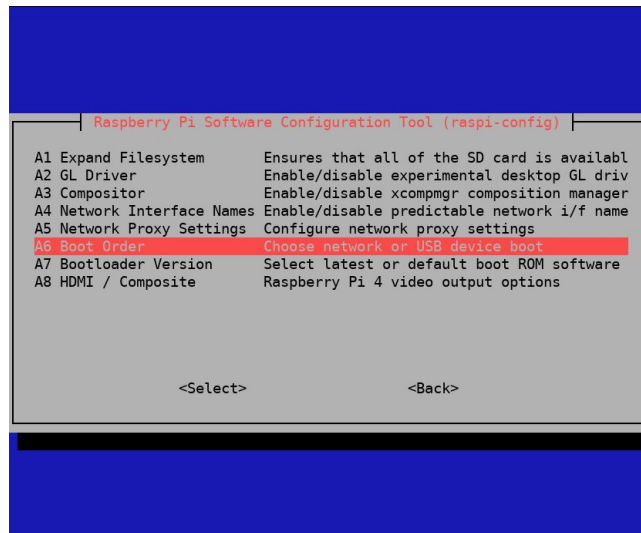


Figure 6: Menu option to select (step 2).

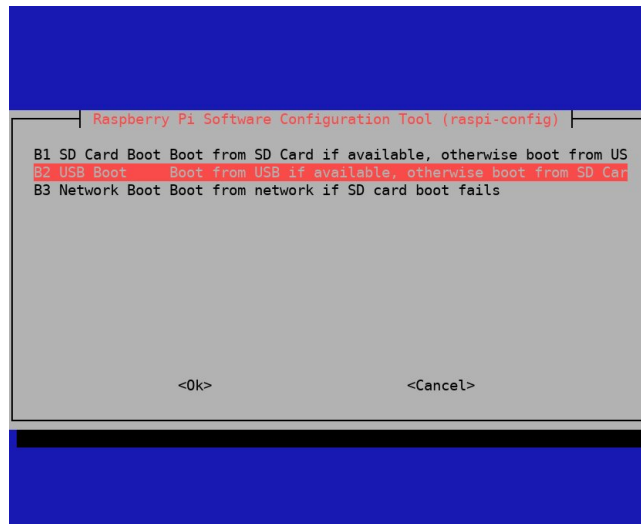


Figure 7: Menu option to select (step 3).

Then just accept the option to reboot the radio. Now the radio is ready to run the HERMES installer. If needed, in order to turn off the radio, type the following in the terminal window (just after raspi-config):

- halt

And after some seconds, when the screen goes black, turn off the radio (using the power key below the power socket).

4 HERMES USB installer preparation

A USB pendrive of at least 8 GB is needed. A computer is needed to prepare the USB pendrive.

The installer should be downloaded from: <https://aco-connexion.org/downloads/installer-image.img.zip>.

After the download is finished, the file should be “unzipped”, as shown in Figures 8, 9 and 10. The extracted file should have the name: “installer-image.img”.

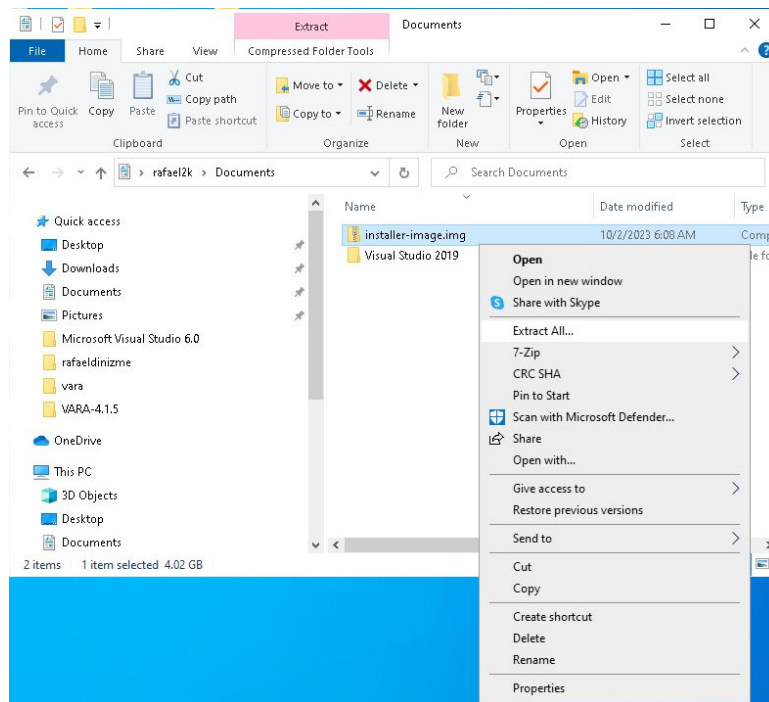


Figure 8: Unzipping the installer file.

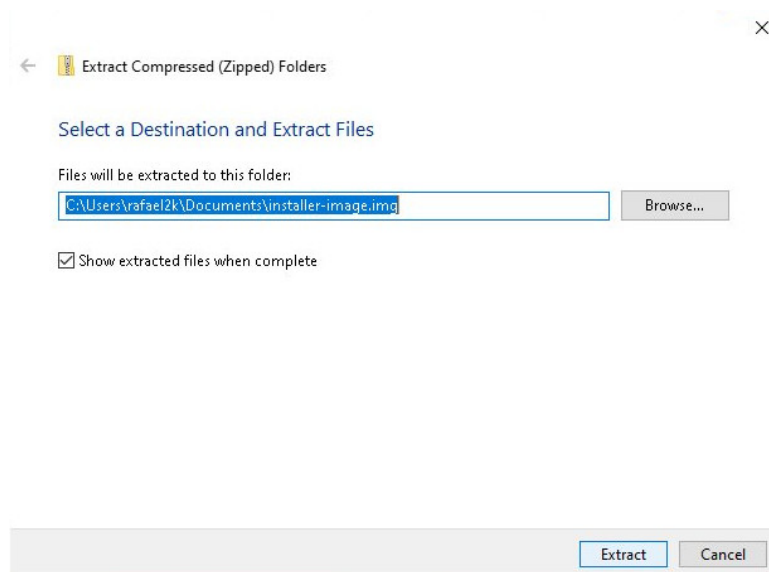


Figure 9: Setting the installer file name.

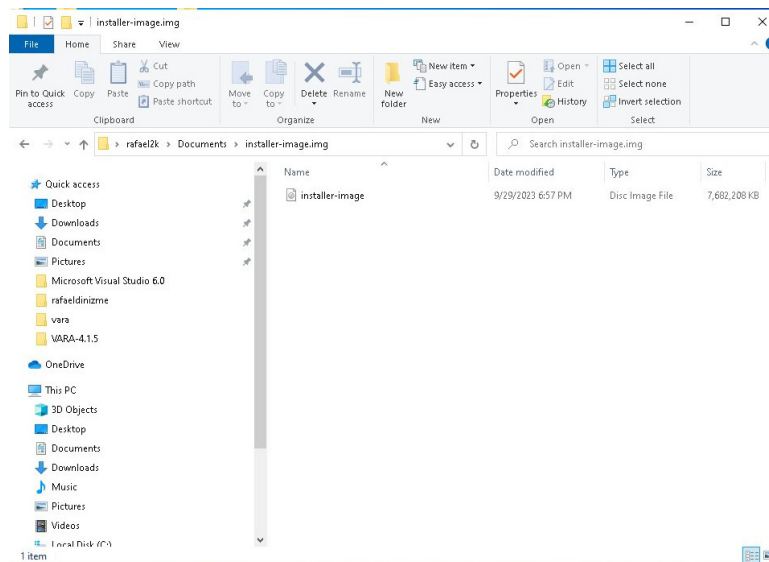


Figure 10: Installer file after unzip.

Then, for copying the installer to the USB pendrive, use the following software (or other raw image writer or flasher), called Etcher (supported on Windows, MacOS and Linux), available in <https://etcher.balena.io/>. The download is available on this website, as shown in Figure 11 (eg: for Windows choose the first option).

DOWNLOAD

Download Etcher

ASSET	OS	ARCH	
ETCHER FOR WINDOWS (X86 X64) (INSTALLER)	WINDOWS	X86 X64	Download
ETCHER FOR WINDOWS (X86 X64) (PORTABLE)	WINDOWS	X86 X64	Download
ETCHER FOR WINDOWS (LEGACY 32 BIT) (X86 X64) (PORTABLE)	WINDOWS	X86 X64	Download
ETCHER FOR MACOS	MACOS	X64	Download
ETCHER FOR LINUX X64 (64-BIT) (APPIMAGE)	LINUX	X64	Download
ETCHER FOR LINUX (LEGACY 32 BIT) (APPIMAGE)	LINUX	X86	Download

Figure 11: Download options for Etcher software.

Run the software Etcher, choose the option “Flash from file” and select the file “installer-image.img”, as shown in Figures 12, 13 and 14.

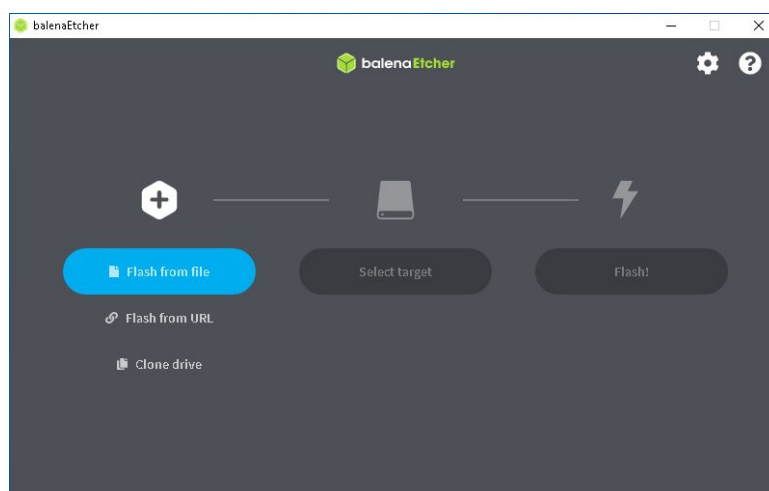


Figure 12: First screen of Etcher, “Flash from file” selected.

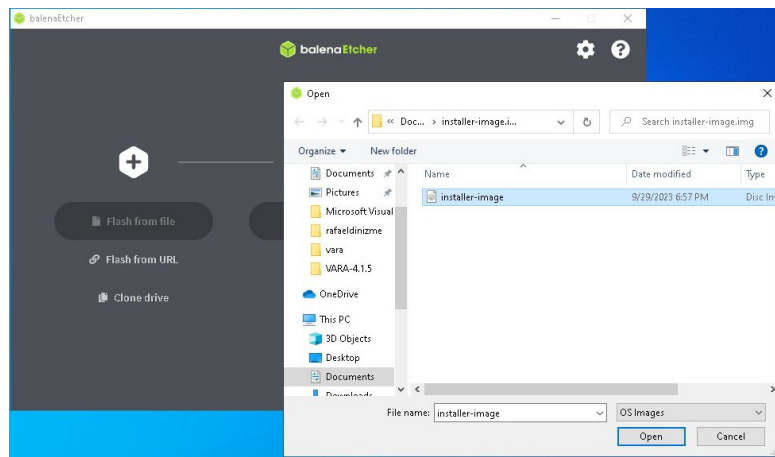


Figure 13: Choose the file “installer-image”.

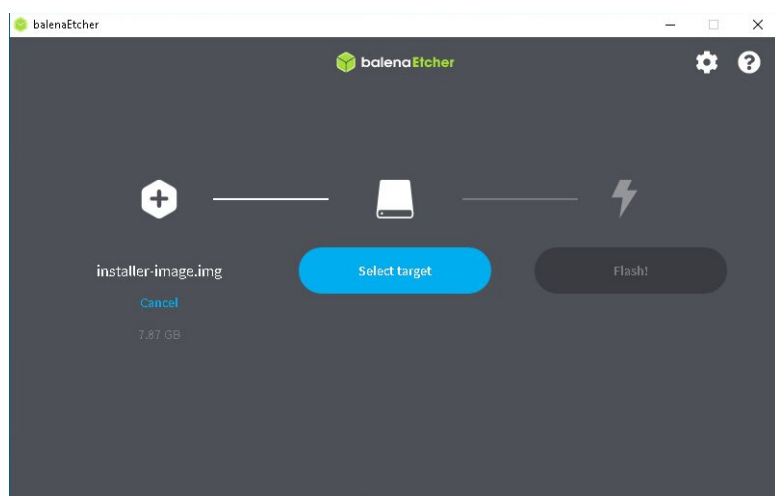


Figure 14: Select the USB pendrive to be used as installer.

After the copy is finished, insert the USB pendrive with the HERMES installer in the radio, as shown in Figure 15. Make sure the radio is turned OFF before inserting the USB pendrive.



Figure 15: Radio with USB pendrive connected, together with keyboard and network cable.

5 HERMES installation

The HERMES installation procedure will set up the sBitx radio with HERMES system, and perform the station-specific setup.

The installation is done in two steps. The first one with the pendrive inserted in the radio, and the second without the pendrive.

After preparing the USB pendrive and plugging it to the radio, turn on the radio.

When asked if you want to proceed with the installation, answer “y” and press ENTER, as shown in Figure 16.

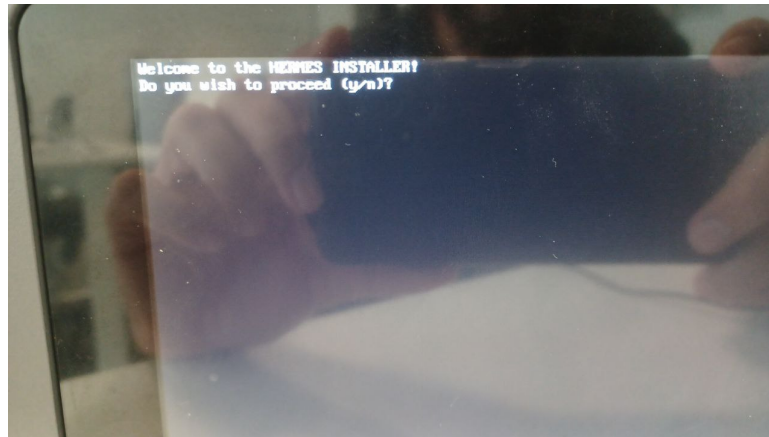


Figure 16: HERMES interactive installer starting.

When asked if you want to write HERMES (copy the system), answer “y” and press ENTER, as shown in Figure 17.

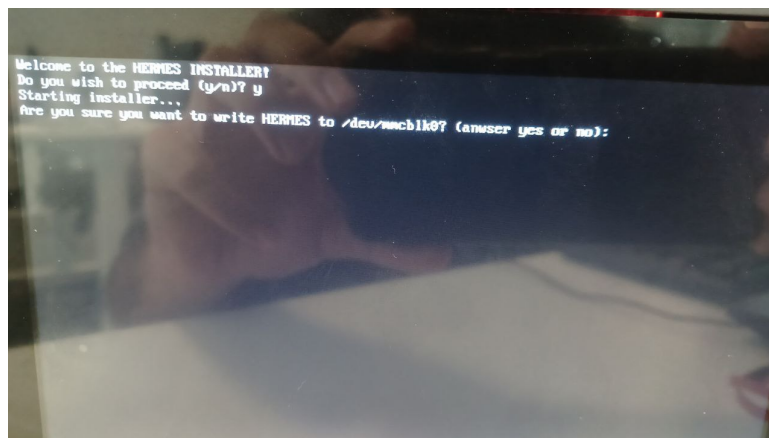


Figure 17: HERMES installer question to start writing to internal radio memory.

Then the installer will copy the HERMES system (Figure 18) to the internal radio SD card, which will take some time, most likely more than one hour.



Figure 18: HERMES installer in process of copying data to internal radio memory.

After the copying is done (shown in Figure 19), reboot the radio (just press any key) and REMOVE the pendrive from the radio for the second step of the installation. Also make sure the network cable (Internet) is connected to the radio.

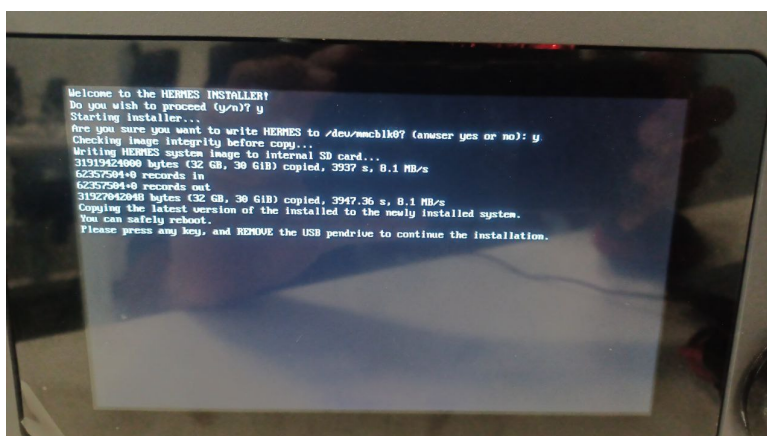


Figure 19: HERMES installer finished copy of the system to internal radio memory.

After the radio restarts, you will be asked about the station name to use, as shown in Figure 20 and 21. Choose one of station names below, by typing the full station name in the installed prompt.

- bakouma.aco-connexion.org
- bangassouhub.aco-connexion.org
- briaahub.aco-connexion.org
- goboudou.aco-connexion.org
- irabanda.aco-connexion.org

The station “bangassouhub.aco-connexion.org” has the gateway station setup. The gateway station should be deployed in a place with Internet connection.

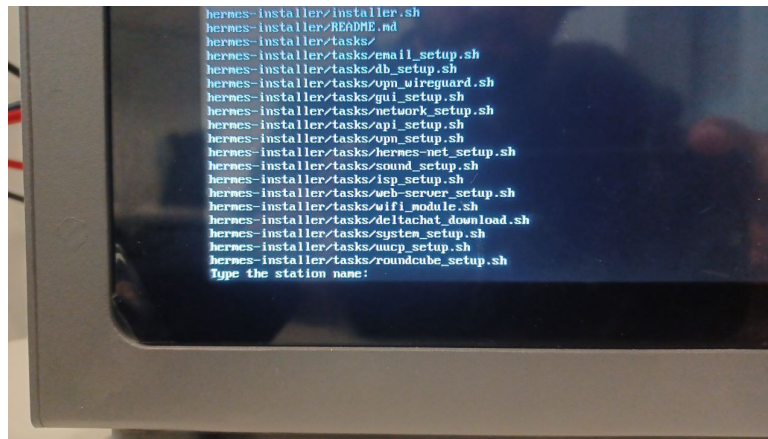


Figure 20: HERMES installer station name prompt.

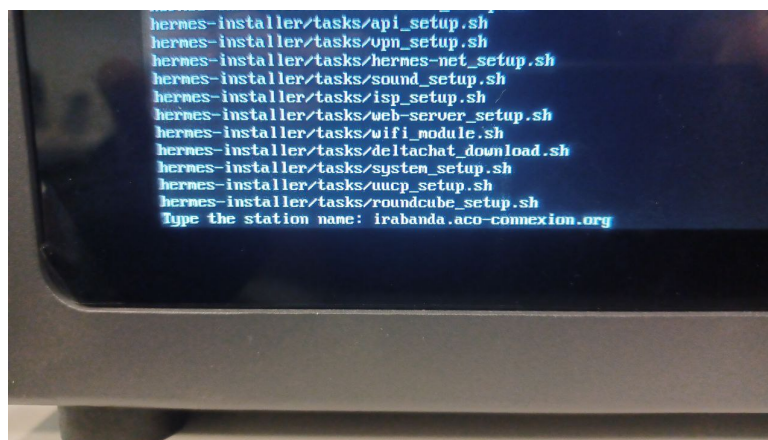


Figure 21: Station name field filled with “irabanda.aco-connexion.org”.

After the setup is finished, you should see the following message “Press any key to REBOOT the radio...”, as shown in Figure 22. Press any key. The installation process is finished.

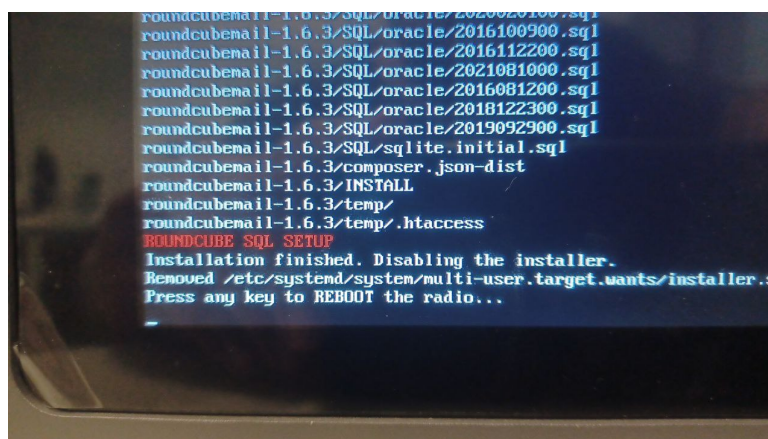


Figure 22: Message that marks the end of installation.

The radio will restart, and the HERMES interface will show up, as shown in Figure 23.

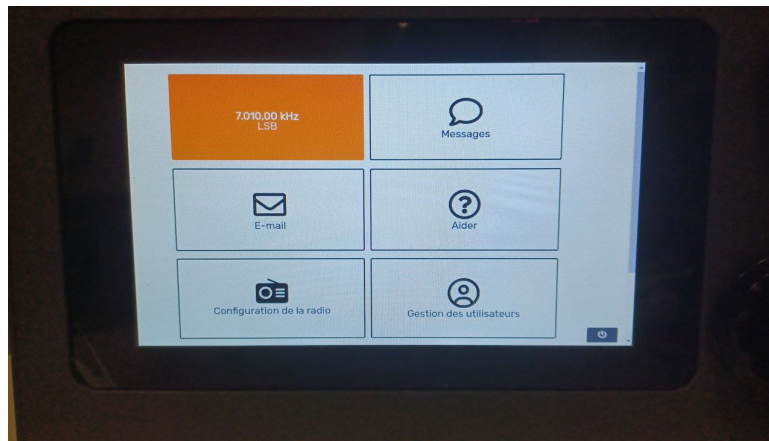


Figure 23: HERMES system screen.

6 WiFi Setup

Default radio's wireless information is:

- Network Name (SSID): hermes
- password: amazonia

In case the system connecting to the wifi of the radio shows a question saying that there is no internet, questioning if you want to stay connected anyway, select the option to continue connected anyway “always” (in case there is also an option to take this action “only once”).

The Wifi settings can be changed through the HERMES interface, as shown in Figures 24 and 25. Choose a network name only with lower case letters and numbers. The password needs to be at least 8 characters, and it is not allowed the use of special characters or spaces - use just lower case letter and numbers. Wifi channel also needs to be chosen, being the allowed channel numbers the range from 1 to 11. Any channel can be chosen. In case two or more HERMES radio are in the same room, choose a different channel and network name for each radio.

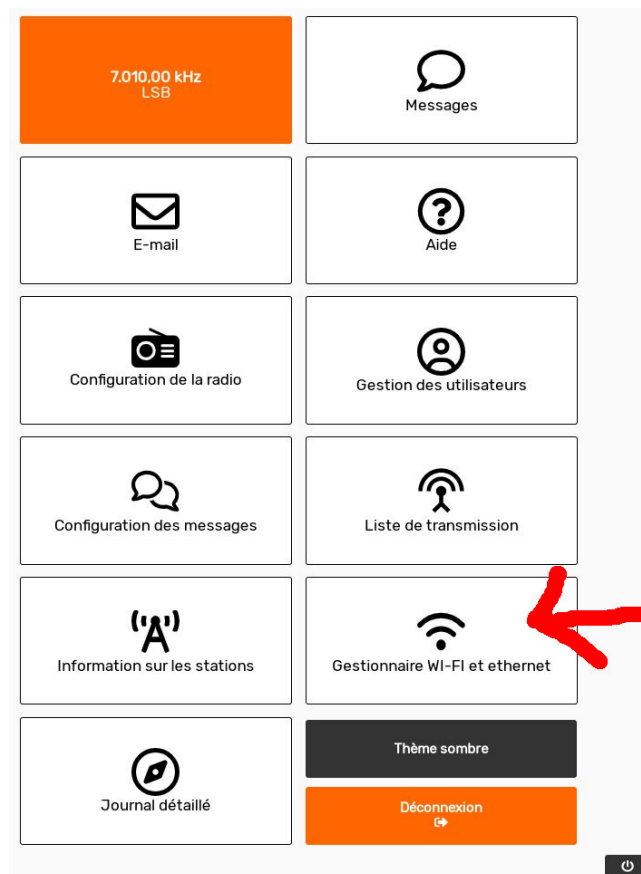


Figure 24: WiFi administrative interface icon pointed by red arrow.

< /home /wifi-manage

Gestionnaire Wi-Fi et ethernet

Nom du réseau SSID

Canal

Mot de passe

Confirmer le mot de passe

sauvegarder les modifications

Figure 25: WiFi administrative interface.

7 Radio Frequency (RF) connections assembly

For proper testing, the radio needs to be connected, ideally, to a power meter (also called wattmeter), which should be connected to a dummy load, as shown in Figures 26 and 27.



Figure 26: Back of the power meter with the cables attached.



Figure 27: Both the dummy load (left) and the power meter (right).

The radio (sBitx) has a BNC female connector, and needs the appropriate cables and adapters to be connected to the Wattmeter and dummy load (eg. by using BNC Male to UHF/SO239 Female). The wattmeter measures the forward and reflected power of the radio, and the dummy load simulates an antenna without radiating the signal.

The radio needs to be connected (see Figure 29) to the “TX” or “Transmitter” (see Figure 28) port of the power meter, while the dummy load should be connected to the “ANT” or “Antenna” port.



Figure 28: Back of the power meter in which the ANT and TX writings can be seen.



Figure 29: The power meter (left) and radio with the connected cable (top right).

Make sure the cables are properly connected and the front of the power meter (Figure 30) is in a visible position.



Figure 30: Front view of a power meter

8 RF test

After all the RF connections ready, a transmit test needs to be made. First, set a frequency in the band you will operate most the radio (Eg. 6200 kHz). Then two tests should be carried:

- Voice test with the microphone, for microphone and its button operation check
- Single tone test, for establishing the power level of the radio

In order to log-in to the HERMES User Interface (UI), a default password is set (that can be changed), and it is:

- user name: root
- password: caduceu

To enter the voice screen in HERMES UI, follow the Figures 31 and 32.



Figure 31: HERMES UI with a finger pointing to the voice functionality icon.



Figure 32: HERMES UI in the voice screen.

The expected result of the voice test, when talking with normal voice level, should be the wattmeter having a oscilating power level, ranging from 0 to 40W.

In order to carry the single tone test, enter the radio configuration screen, as shown in Figure 33. The single tone functionality is enabled using the “PTT” button in the radio configuration page, as shown in Figure 34.

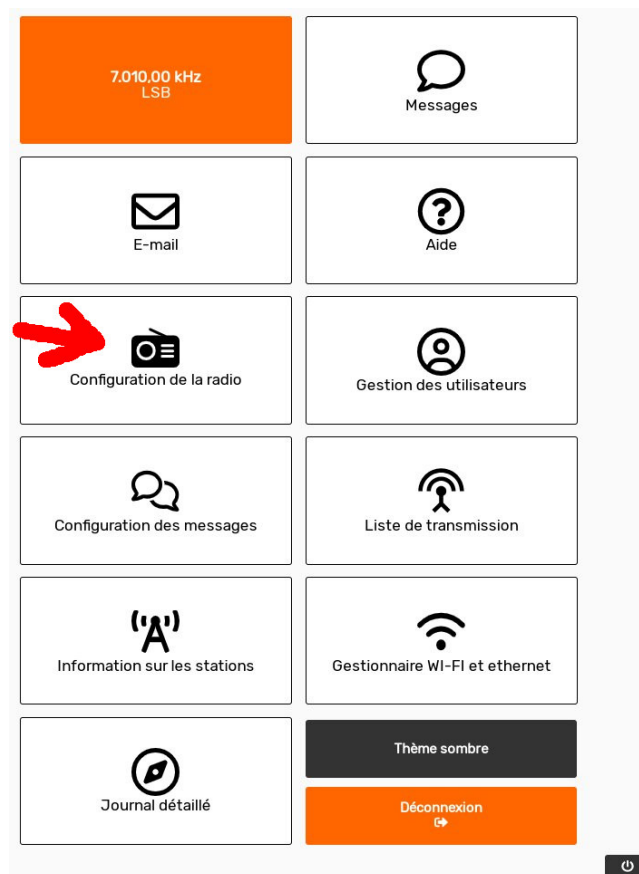


Figure 33: HERMES UI with an arrow pointing to the radio configuration icon.

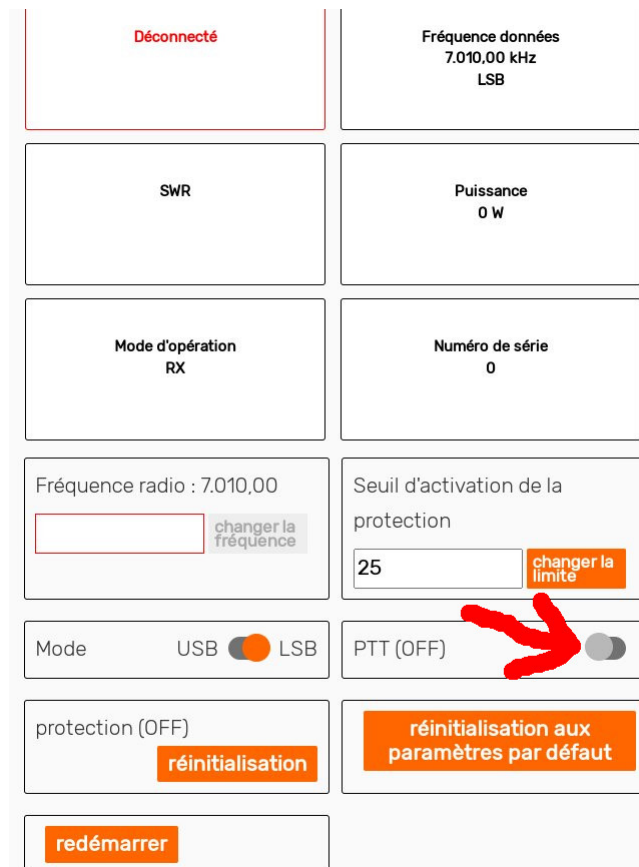


Figure 34: HERMES UI in the radio configuration page, an arrow points to the “PTT” button.

The expected result of the single tone test, should be a power reading of approximately 20 W, as shown in Figures 35 and 36.

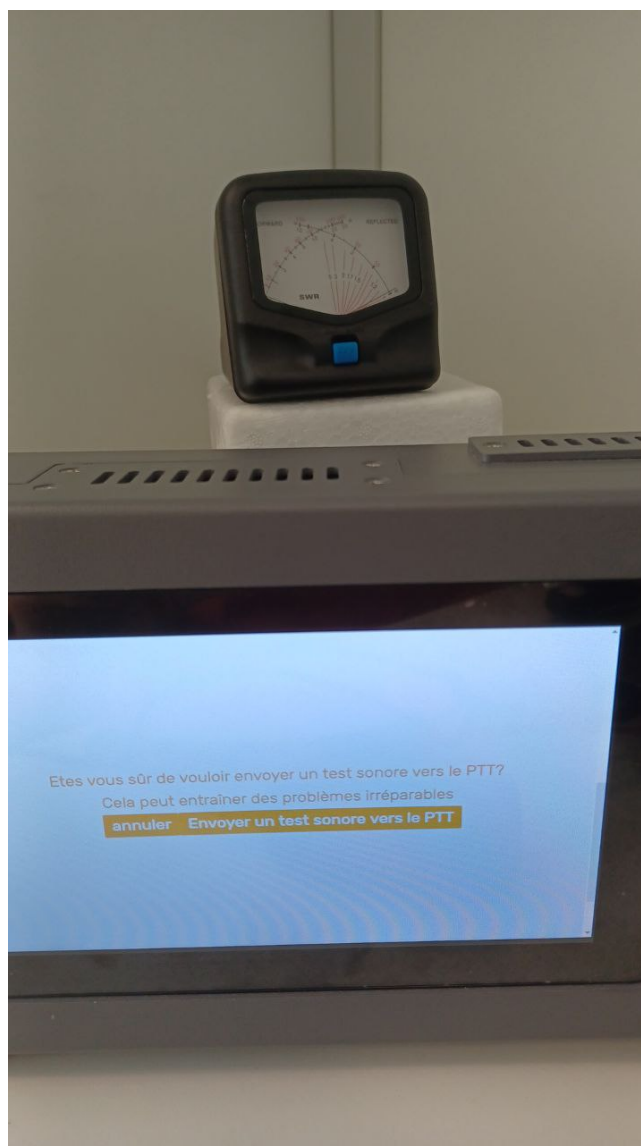


Figure 35: Confirmation question shown after clicking to enable the PTT. No power.

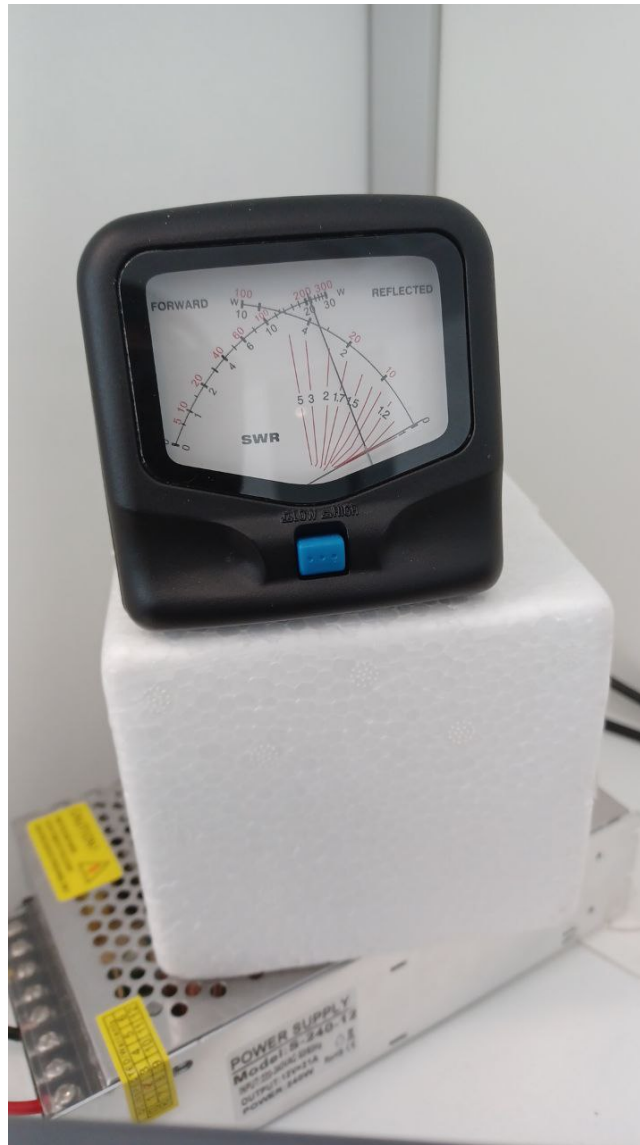


Figure 36: Wattmeter showing 20W of power.

9 E-mail setup and test

In this section the process of creating one or more users in the HERMES system is address. Also the process of connecting a device (eg. tablet or cellphone) to the radio over WiFi and using the software DeltaChat (an email client) for messages exchange.

In the HERMES interface, create one or more users, to be used for the email access. Then connect the companion device to the radio WiFi network, and install the software DeltaChat (an email client). Follow the steps shown in Figures 37 and 38 in order to create users in the HERMES UI.

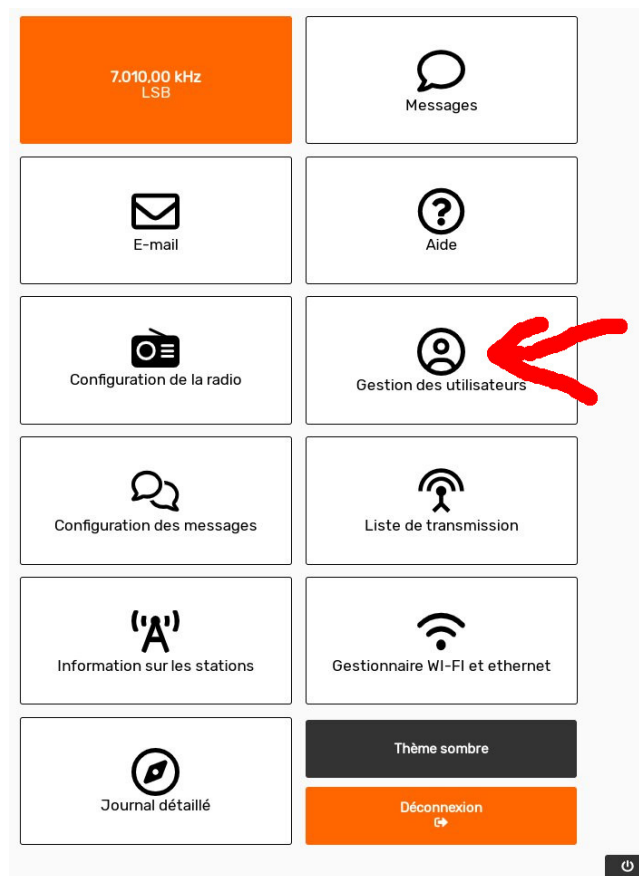


Figure 37: User management icon in main HERMES page.



Figure 38: User management icon in main HERMES page.

The in the device, first connect it to the WiFi radio network, as described in section 6. Then open the DeltaChat application, which available in any app store, or locally in the radio, as shown in Figure 39 and the, by clicking the DeltaChat icon, download links will appear, as shown in Figure 40.

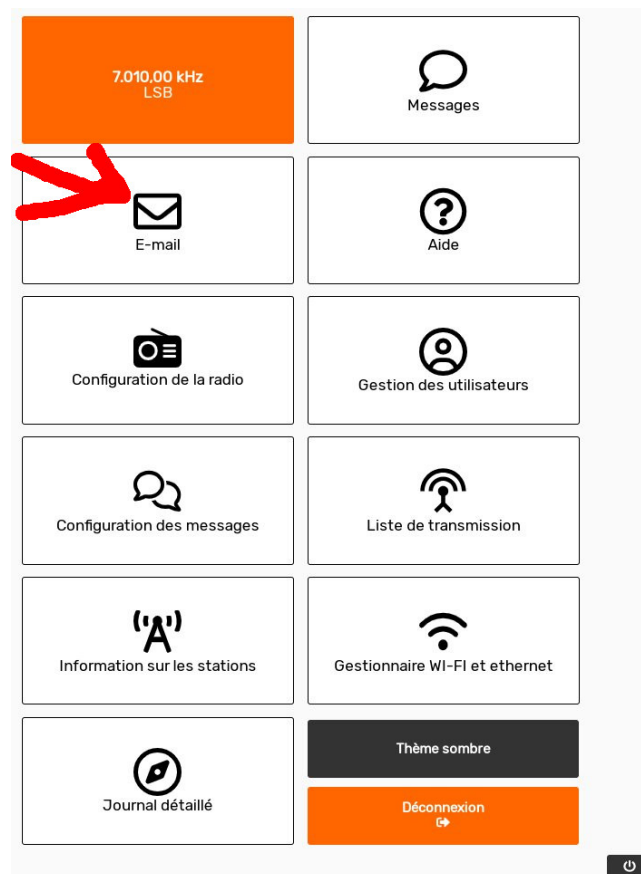


Figure 39: User management icon in main HERMES page.

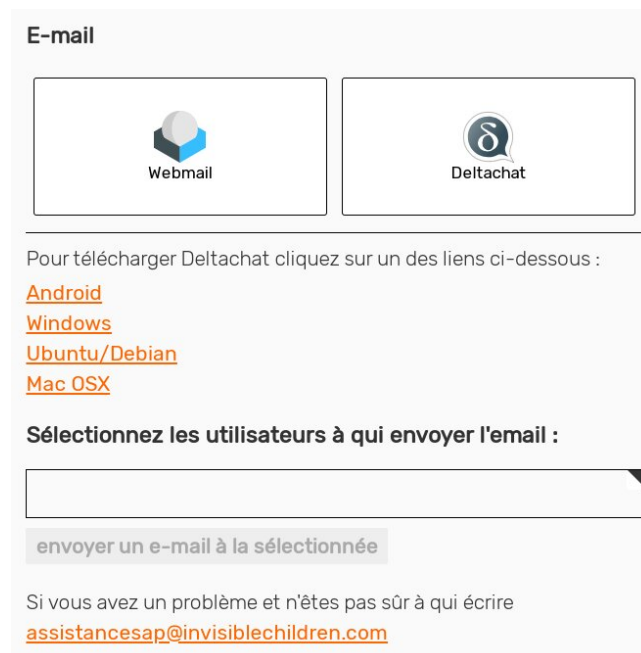


Figure 40: User management icon in main HERMES page.

In DeltaChat, go in “Access your server” and put the email and password created in HERMES system (see Figure 41). The email should look like “opérateur@irabanda.aco-connexion.org”, and is composed by the HERMES username, an “@”, and the name of the station.

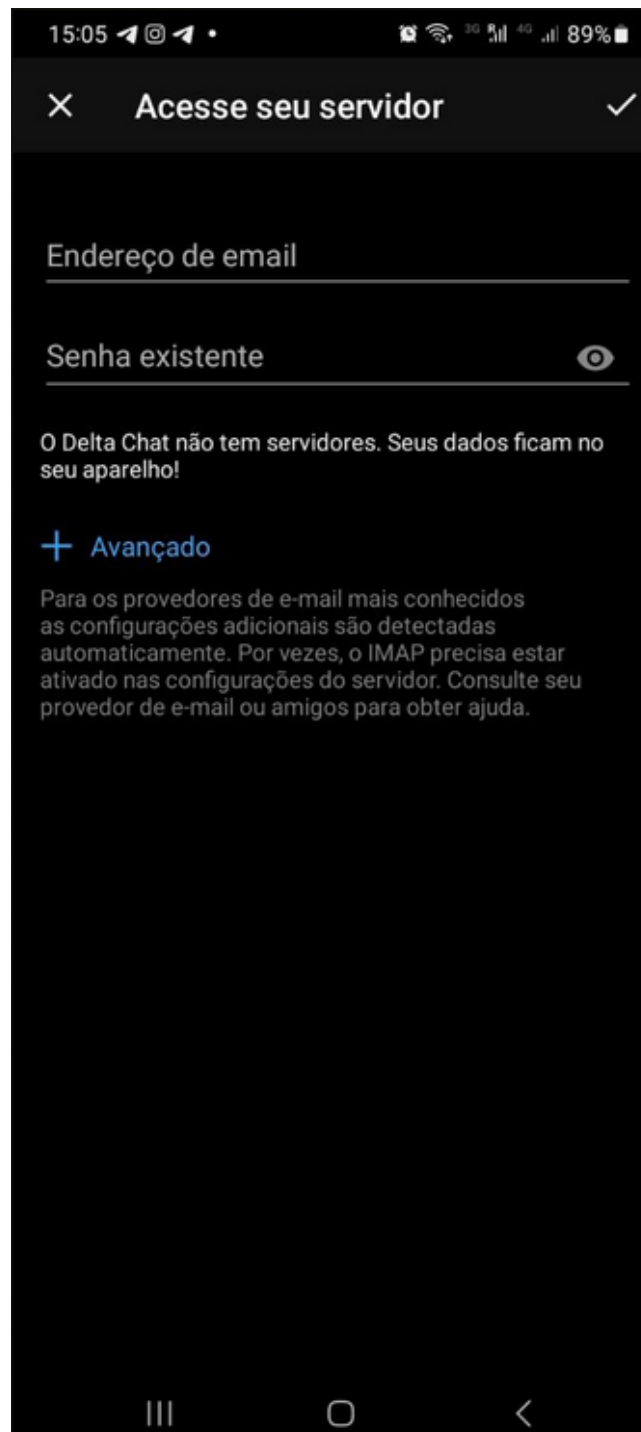


Figure 41: DeltaChat login screen, with email and password fields shown.

10 Further troubleshooting (maintainers)

This installation guide focuses on hardware setup and basic functional tests. For a detailed HERMES system architecture and software stack description (radio control via `sbitx_controller`, modem/TNC integration, `uucpd/uuport`, and gateway scheduling), see the document “**HERMES Software Description**” (`software_en.pdf`) in the HERMES documentation repository.

Useful service-level commands:

```
systemctl status sbitx uucpd
```



```
journalctl -f -u sbitx  
journalctl -f -u uucpd
```